MV-MoE: A Multiview Expert Fusion Framework for Multispectral Object Detection

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***Abstract*—Multimodal image fusion, particularly the collaborative modeling of infrared and visible light images, plays a crucial role in tasks such as nighttime target perception and visual enhancement. Although existing methods have enhanced the expressive capabilities of multimodal information fusion to some extent, they struggle to fully construct the multiperspective semantic structures inherent in different modalities. This results in a lack of controllability in fusion pathways, insufficient information coordination, and a high susceptibility to semantic conflicts. To address this, we propose a fusion framework of a mixture of experts based on multiview parallel expert groups (MV-MoE). This framework enables path modeling and expert activation scheduling within a hierarchical structure. The method establishes multiperspective semantic expert groups driven by a hierarchical dynamic gating mechanism, enabling perspective selection and adaptive data fusion during feature extraction. The core approach involves concatenating multisource data features and feeding them into a hierarchical parallel expert network to extract multiview feature information. A dynamic gating mechanism then weights and fuses feature results from different perspectives, enabling multiview feature extraction and dynamic fusion of multimodal data. Building upon this foundation, an intragroup collaborative regularization strategy is introduced to enhance coordination and generalization capabilities among expert networks, ensuring stability and reliability in fusion. Concurrently, a globally shared expert module is incorporated to balance multiview detail information with holistic context, further improving consistency in the fusion process and stability in object detection. Extensive experimental validation across multiple datasets conclusively demonstrates the effectiveness of the proposed approach. The code is available at https://github.com/AnWiLL-Work/MV-Moe.git**

***Index Terms*—Mixture of experts (MoE), multimodal, multispectral object detection, multiview information.**

I. Introduction

INFRARED and visible light image fusion is a key multimodal perception technology with extensive applications in practical scenarios such as intelligent security, nighttime surveillance, target detection, and tracking [1], [2]. RGB and IR images each possess distinct advantages [3], [4]. In well-lit daytime environments, RGB enhances details such as building edges, wheel outlines, and grass textures. Conversely, in nighttime or low-light conditions, IR images accentuate thermal sources like pedestrians and vehicle engines [5], [6]. Despite their high complementarity in perceptual dimensions, significant differences in data distribution, structural features, and signal intensity pose substantial challenges for efficiently fusing IR and RGB images. Key unresolved issues include modeling modal redundancy and complementarity, avoiding information conflicts and ambiguities during fusion, and achieving semantically consistent and detail-aligned fusion representations.

In recent years, image fusion methods based on convolutional neural networks [7] and transformers [8], [9] have made significant progress. A series of approaches have attempted to effectively combine information from two modalities through attention mechanisms [10], generative adversarial networks (GAN) [11], [12], or feature reconstruction strategies, thereby enhancing the visual quality of fused images and improving performance in downstream tasks. However, existing fusion methods lack decoupled modeling for multimodal fusion data. Directly blending modal features fails to address variations in local modal importance, leading to image information interference [13]. For instance, in regions where object textures are already clear, supplementing with IR images may cause information overlap, creating conflicts that blur the fused object.

In multimodal data fusion, information from different modalities possesses distinct characteristics and advantages [14], [15]. Learning processes must fully consider the unique information provided by each modality and adaptively select the most suitable feature fusion strategy based on the data's properties [16], [17]. Therefore, we propose a fusion module capable of dynamically extracting features from different perspectives based on modal characteristics to address information conflicts encountered in multimodal fusion. Our method distinguishes the primary and secondary information during the fusion process at the structural level, thereby achieving more precise and efficient multimodal data fusion. Specifically, we have implemented a dynamic fusion framework named the multiviewed mixture of experts (MV-MoE) module, which integrates a multimodal fusion gating mecha-

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nism to guide the collaboration between the globally shared expert groups and the parallel expert groups. Additionally, we introduce an expert regularization module to maintain balance among the expert groups.

In the hierarchical parallel expert network, we employ a learnable dynamic gating mechanism that integrates input features to guide stacked expert networks in generating multiview information through feature fusion. Expert weights are computed based on gating calculations, enabling multiview feature extraction and dynamic fusion of multimodal data. Furthermore, we employ intragroup expert collaborative regularization strategies to stabilize expert allocation within parallel expert groups. This enhances coordination and generalization capabilities among expert networks, promotes consistency in expert outputs, and ensures stable and reliable fusion. Additionally, we introduce a global shared expert module to extract coarse-grained information from the fused image. This serves as a global semantic baseline for all modalities and expert groups, balancing multiview detail and overall context. This module further enhances consistency in the fusion process and improves object detection stability. By integrating global holistic information with parallel multiview information, our model effectively fuses image information from different modalities.

The main contributions of this article are as follows.

1) We propose a dynamic fusion framework based on the mixture of experts (MoE), aiming to provide an adaptive solution for multimodal image fusion that optimally blends global information and multiview details. This approach dynamically extracts and integrates information from different perspectives within the fused image.
2) We design a hierarchical feature-selection fusion strategy that enables the model to progressively choose and utilize the most relevant modality information based on feature importance during the fusion process. This strategy structurally reduces cross-modal interference and leads to more stable fusion results in both local details and overall representation.
3) We introduce an intragroup collaborative regularization method based on the characteristics of parallel expert groups. This approach effectively adjusts the outputs of expert groups, enabling the model to comprehensively integrate and learn information from multiple perspectives within the fused features.

II. Related Work

A. Infrared and Visible Light Image Fusion

Infrared and RGB image fusion is a critical multimodal perception task [18], [19], with its core challenge lying in balancing semantic information enhancement and fine-grained feature preservation in the fused image. Early methods primarily focused on fusion strategies between the pixel layer and the transform domain, such as weighted averaging, principal component analysis (PCA), nonsubsampled contourlet transform (NSCT), and wavelet transforms [20], [21]. However, these methods typically relied on handcrafted features and lacked deep semantic modeling capabilities. In recent years, deep learning approaches have gained prominence [22], [23], including CNN, attention mechanisms, and GAN. Hui Li effectively preserved intermediate layer features by combining traditional CNN layers with dense connection blocks, optimizing information and gradient flow by dense connections [24]. Ma et al. [12] use the adversarial interaction between the generator and discriminator. While the generator preserves the primary intensity of IR images and the additional gradient information from RGB images, the discriminator compels the fused image to incorporate more details from the images. Xu et al. [25] employed feature extraction and information measurement to automatically estimate the importance of source images. Based on this, they determined an adaptive degree of information retention to preserve similarity between the fusion result and the source images. These methods significantly enhance the clarity and interpretability of fused images [26], [27]. However, such approaches often fully fuse both modalities at an early stage, lacking structural decoupling for different modal perspectives and failing to dynamically select fusion strategies. This results in blurring or instability during target edge detection and heat source identification. We propose a dynamic fusion framework that provides adaptive solutions to optimize the integration of global information and multiperspective details. By flexibly extracting information from images and effectively integrating details from different viewpoints, it achieves dynamic fusion of multimodal images.

B. MoE Model

The MoE architecture has demonstrated formidable modeling capabilities in recent years as a parameter-scalable, sparsely inferable network structure across tasks such as large-scale language modeling, image classification, and multitask learning [28], [29]. Notably, several studies have leveraged MoE in tackling multimodal tasks. For instance, Riquelme et al. [30] developed token-processing experts and integrated the VisionTransformer to obtain variable-path representations, thereby boosting the flexibility and adaptability of image representations. Moreover, Sun et al. [31] put forward an innovative dynamic image fusion framework that dynamically extracts effective and comprehensive information by steering hybrid local and global experts via multimodal gating, consequently enriching the overall texture details and contrast of the fused images. Zhu et al. [32] proposed task-customized mixture of adapters (TC-MoA), which views each expert as a fine-tuning adapter. These adapters are shared among diverse tasks and are subject to mutual information regularization, guaranteeing task compatibility and multisource image complementarity. Xin et al. [33] introduced weakly supervised interaction loss to design multiple experts for multimodal learning, enabling a data-driven interaction mechanism. They employ a reweighted model that assigns importance scores to each interaction expert's output, ensuring sample-level and dataset-level interpretability. Although MoE offers expressive flexibility for image fusion tasks, existing methods commonly suffer from issues such as monotonous structural organization, uneven expert utilization, and shallow fusion pathways [34], [35]. Most approaches employ a unified expert pool without establishing a modality-structured expert grouping mechanism

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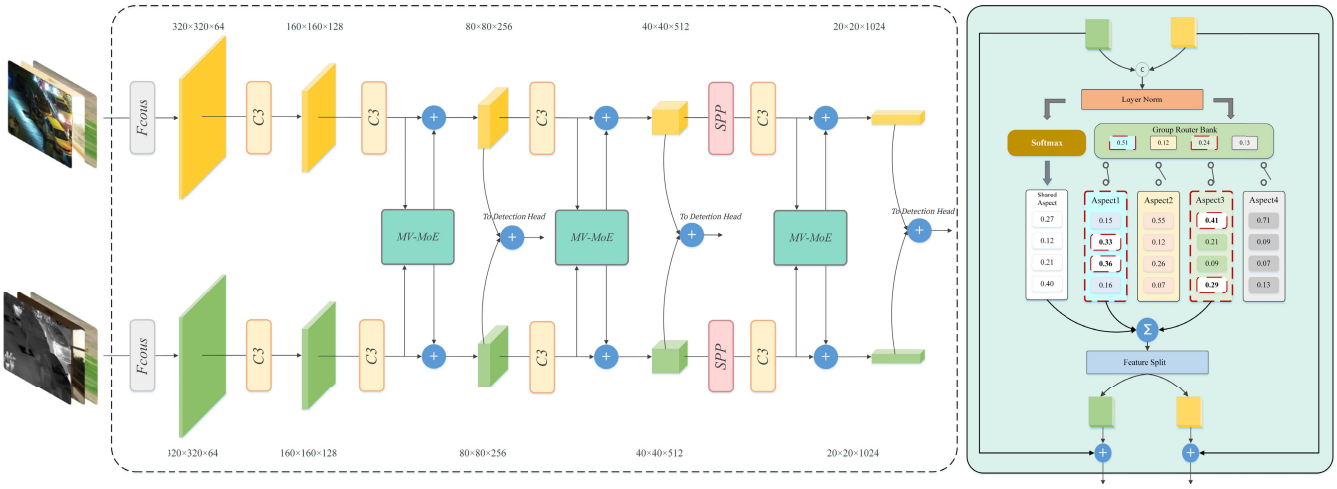


Fig. 1. Proposed fusion network architecture. This framework mainly comprises a dual-branch feature extraction network and three multiview MoE fusion modules. C3 and SPP denote the feature extraction ensemble module and spatial pyramid pooling module of YOLOv5, respectively. The right side of the figure illustrates the internal structure of the multiview MoE fusion module. Finally, the optimized features from the two branches are aggregated using a "+" operation and fed into three detection heads of different scales for multiscale object prediction.

[36], [37], [38]. Additionally, the absence of collaborative constraints and shared pathways among experts leads to information overlap and conflicts during semantic fusion. To address this, our proposed method meticulously models the complex multiview semantic combination relationships that exist between IR and RGB images. This approach achieves refined extraction of multimodal features by constructing multiple semantic subspaces for input features, effectively reducing redundant information and thereby enhancing the accuracy and efficiency of multimodal fusion.

III. Method

A. Overall Architecture

We propose MV-MoE, a multimodal image fusion framework that leverages parallel expert groups and hierarchical gating mechanisms. This framework is designed to overcome the common issues in existing fusion methods, such as flat expert structures, coupled modal paths, semantic conflicts, and expressive redundancy. Structurally, MV-MoE divides experts into multiple subexpert groups. With regard to cross-modal features $X \in \mathbb{R}^{H\times W\times 3}$, it dynamically selects fusion paths via two-level gating mechanisms, namely intergroup (which performs macroscopic routing to select the most suitable semantic expert groups) and intragroup (which performs fine-grained selection of individual experts within the activated groups). Simultaneously, it introduces a shared expert module for global semantic completion of fused image information and designs a collaborative regularization term to enhance cooperation and balanced utilization among experts. As shown in Fig. 1, we first employ a dual-branch YOLOv5 network [7] to extract features from input RGB images $X_r \in \mathbb{R}^{H\times W\times C}$ and IR images $X_i \in \mathbb{R}^{H\times W\times C}$, respectively. This network captures multiscale image features, with the C3 module connecting multiple downsampling convolutional layers via residual connections for efficient information extraction. Subsequently, we integrate cross-modal features from intermediate layers via incorporated MV-MoE layers, as shown in Fig. 2. The data processing within this module follows a clear four-stage pipeline. Following the initial feature extraction, the hierarchical gating mechanism computes routing weights via adaptive gating and normalization gating layers. Crucially, in the expert groups stage, rather than deeply decoupling target features from complex backgrounds, the heterogeneous experts decouple the fusion features into multiple distinct orthogonal directions in the feature space, ensuring comprehensive multiview semantic extraction. Finally, dynamic fusion aggregates these directional features. Concurrently, the computed gating weights are used to calculate the collaborative and load-balancing regularization losses, seamlessly integrating the routing mechanism with the overall training objective. Through these MV-MoE layers, our network progressively integrates relevant information.

B. Dynamic Gating Mechanism and Parallel Expert Groups

1) Group-Level Gating: Traditional MoE architectures primarily rely on a unified expert pool for selection, disregarding multiperspective information at the semantic level. To alleviate this issue, building upon the standard noisy Top-K gating mechanism [39], we introduce a tailored variant to adapt to our multimodal routing requirements. Specifically, we propose a hierarchical solution by incorporating an expert grouping mechanism that divides experts into G groups, with each group symbolizing a specific semantic perspective. We enhance model robustness and path diversity through the implementation of a linear gating mapping and learnable noise perturbations. Input features $x \in \mathbb{R}^{N\times C}$ traverse this group-level gating module to calculate activation scores for each expert group

$$s^{\text{group}} = Softmax\Big(TopK_g\Big(X\cdot W_g + Softplus\left(X\cdot N_g\right)+\varepsilon_g\Big)\Big), \varepsilon_g \sim N\left(0,\sigma^2\right) \tag{1}$$

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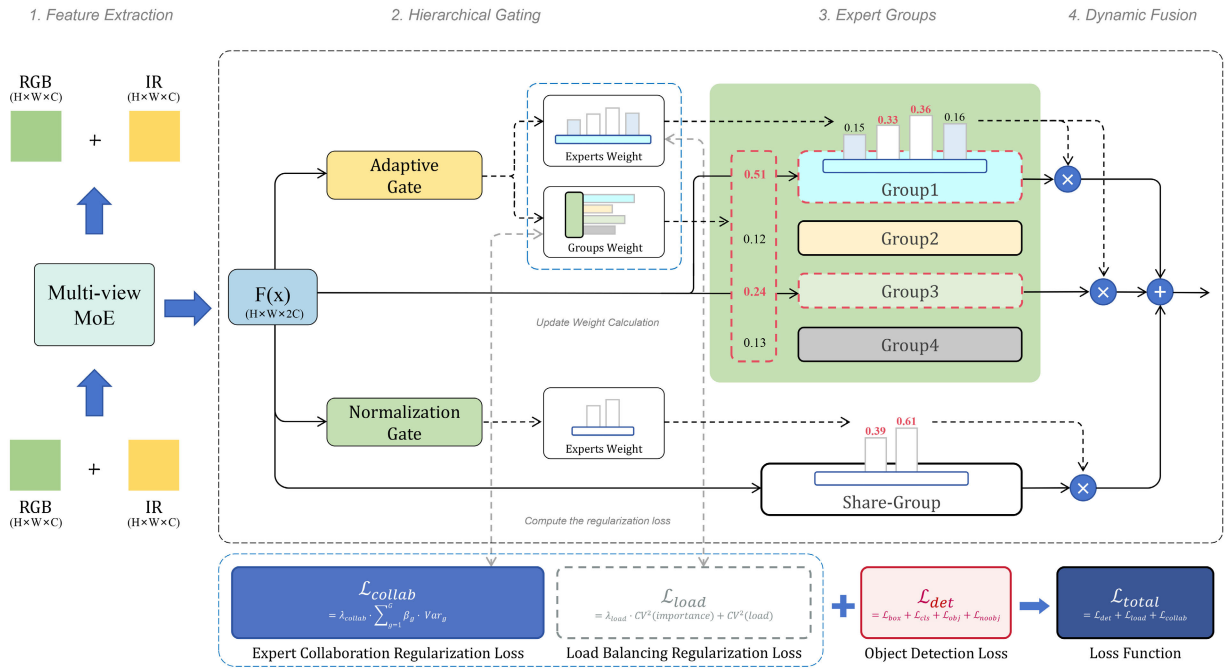


Fig. 2. Architecture of the MV-MoE module. The input features F(x) are processed through two distinct paths. Solid arrows represent the data flow, which feeds the features into the multiview expert group and the shared expert group for feature extraction, and then into the adaptive gating and normalization gating layers to compute the activation weights for each expert group and expert. The dashed arrows represent the weight computation flow, where the activation weights output by the adaptive gating and normalized gating are multiplied by the corresponding expert outputs, summed, and aggregated to produce the final fused features. The bottom panel illustrates the computation of regularization losses ($\mathcal{L}_{collab}$ and $\mathcal{L}_{load}$) derived from gating weights to form the total loss ($\mathcal{L}_{total}$).

where $W_g \in \mathbb{R}^{C\times G}$ is the group gating parameter matrix for the task, N_g is the noise matrix for groups gating, and ε_g is Gaussian noise used to enhance model robustness and experts diversity. Subsequently, $\mathrm{TopK_g}(\cdot)$ selects the top K_g expert groups with the highest weights based on the final scores from group-expert gating. These groups are activated and participate in the next layer selection.

2) Expert-Level Gating: Within each selected expert group g, an intragroup gating mechanism further selects K_e experts from the group's E experts. Input features x are similarly scored via intragroup gating parameters and noise perturbations

$$s_g^{\text{expert}} = Softmax\Big(TopK_e\Big(X\cdot W_e + \big(Softplus\big(X\cdot N_e\big)+\varepsilon_e\big)\Big), \varepsilon_e \sim N\big(0,\sigma^2\big)\Big) \tag{2}$$

where $W_e \in \mathbb{R}^{C\times E}$ is the intragroup gating parameter matrix, N_e is the group gating noise matrix, and ε_e is Gaussian noise. Finally, the top K_e experts from each activated group are selected for activation and feature extraction.

3) Expert Fusion Output: Combining intergroup and intragroup gating weights, the model eventually channels the input features x into the chosen K_g experts within the group. The outputs of these experts are then fused through a weighted-merging process, culminating in the formation of the final multiperspective output

$$y_{multiview} = \sum_{g\in G'} \beta_g \cdot \left(\sum_{i\in E'_g} \alpha_{g,i}\cdot f_{g,i}(x)\right) \tag{3}$$

where β_g represents the intergroup gating score, $\alpha_{g,i}$ denotes the gating weight of the ith expert within the group, $f_{g,i}(\cdot)$ is the output of the ith expert in the gth group, while G' and E'_g denote the selected expert groups index set and the experts index set within the group, respectively.

C. Shared Expert Layer

To enhance the sharing and fusion of multimodal information, we introduce an independent expert module outside the parallel expert groups as a shared expert pool. Comprising S experts, it assists in globally reinforcing the semantic output of parallel expert groups. Unlike structurally organized expert groups, the shared expert pool provides the same common semantic mapping for all inputs. The combination method for shared experts is learned through a fully connected layer. Its output undergoes normalization to serve as the weighting coefficients for shared experts

$$y_{shared} = \sum_{s=1}^{S} Softmax(W_s x)\cdot f_{shared}(x). \tag{4}$$

where $f_{shared}(\cdot)$ represents the expert's output result. The input feature x, after passing through the linear output and normalization functions, serves as the expert weight. The

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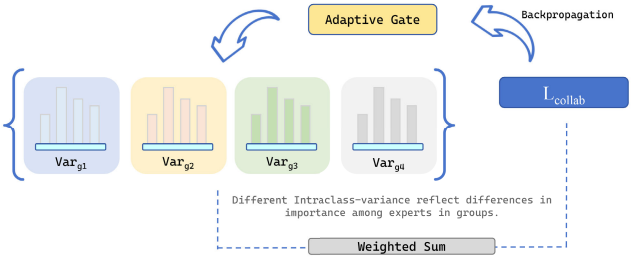


Fig. 3. Collaborative regularization loss.

output of the shared expert layer is derived through a weighted summation of the outcomes generated within the shared expert group. This is subsequently integrated with the output from the parallel expert group to produce the final result

$$y_{final} = y_{multiview} + \lambda \cdot y_{shared} \tag{5}$$

where $\lambda \in [0, 1]$ controls the weighting ratio between the two components, with a default value of 0.5 in experiments.

D. Expert Collaboration Regularization and Load Balancing

To further enhance semantic expression stability within parallel expert groups, we designed an intragroup expert collaboration regularization term as illustrated in Fig. 3. This term constrains the consistency of weight distribution among experts within each group. It encourages experts in the same group to achieve similar gate activation when processing identical features, thereby improving model stability and interpretability in expert selection. Specifically, to ensure that the expert groups maintain consistent activation responses when encountering similar semantic features, let the gating weight of the ith expert in group g be $\alpha_{g,i}$, and the group average weight be $\bar{\alpha}_g$. The intragroup weight variance is computed to measure this activation consistency

$$\mathrm{Var_g} = \frac{1}{E}\sum_{i=1}^{E}\left(\alpha_{g,i} - \bar{\alpha}_g\right)^2. \tag{6}$$

We further use the intergroup gating score β_g to weight the variance of each group and employ λ_{collab} as the hyperparameter for the collaborative regularization term, yielding the collaborative regularization loss

$$\mathcal{L}_{collab} = \lambda_{collab} \cdot \sum_{g=1}^{G} \beta_g \cdot \mathrm{Var_g}. \tag{7}$$

To prevent certain experts from remaining inactive for extended periods (i.e., the "expert collapse" phenomenon), which leads to a reduction in experts' concentration and generalization abilities, we also employ a regularization term based on the coefficient of variation (CV^2) [17], [39] to encourage expert load balancing

$$\mathrm{CV}^2(x) = \frac{\mathrm{Var}(x)}{(\mathrm{E}[x])^2 + \varepsilon} \tag{8}$$

$$\mathcal{L}_{load} = \lambda_{load} \cdot \mathrm{CV}^2(importance) + \mathrm{CV}^2(load) \tag{9}$$

where $\mathrm{CV}^2(x)$ is the coefficient of variation regularization term, *importance* denotes the weighted sum of all experts activated during training, *load* represents the frequency each expert is selected under noisy perturbations, and λ_{load} is the coefficient adjusting the regularization strength. The final total loss function is updated to [40], [41], [16], and [18]

$$\mathcal{L}_{det} = \mathcal{L}_{box} + \mathcal{L}_{cls} + \mathcal{L}_{obj} + \mathcal{L}_{noobj} \tag{10}$$

$$\mathcal{L}_{total} = \mathcal{L}_{det} + \mathcal{L}_{load} + \mathcal{L}_{collab} \tag{11}$$

where $\mathcal{L}_{det}$ is the composite loss function for object detection in YOLOv5, $\mathcal{L}_{box}$ represents the localization error between predicted and ground truth bounding boxes, $\mathcal{L}_{cls}$ is the category classification loss based on binary cross-entropy, and $\mathcal{L}_{obj}$ and $\mathcal{L}_{noobj}$ are the confidence prediction losses for positive and negative sample locations, respectively. Additionally, $\mathcal{L}_{collab}$ and $\mathcal{L}_{load}$ control the weight hyperparameters for collaborative regularization and load balancing regularization, respectively, with default settings of 0.001 and 0.01 in this article.

IV. Experiments

A. Dataset

To validate model performance, we selected the LLVIP, VEDAI, and M3FD datasets for experimentation. The LLVIP dataset, which includes RGB and IR image fusion of pedestrians under low-light conditions, assesses feature extraction and object detection capabilities in complex lighting scenarios. The VEDAI dataset, focused on vehicle detection in aerial imagery, tests the model's generalization ability in high-resolution, complex backgrounds. The M3FD dataset, comprising multimodal in-vehicle footage from urban road environments, evaluates the model's multimodal data fusion and object detection capabilities in real-world traffic scenarios. Collectively, these datasets represent diverse application scenarios and challenges, effectively validating the model's comprehensive performance in multimodal data fusion, object detection, and feature extraction. Details of these datasets are provided below.

1) LLVIP [42]: A low-light vision dataset was specifically constructed by Beijing University of Posts and Telecommunications for nighttime pedestrian detection and image enhancement tasks. The dataset comprises 15 488 pairs of aligned RGB and IR images, capturing 24 nighttime/low-light scenes and two daytime scenes. It serves as an excellent resource for validating the robustness of fusion strategies in challenging low-light conditions and multiobject interference scenarios.

2) VEDAI [43]: This dataset is specifically designed for vehicle detection and object recognition tasks on high-resolution remote sensing imagery. Based on aerial remote sensing images, it covers diverse road traffic and urban building scenarios. The images contain numerous small-sized vehicle targets, including nine categories such as cars, trucks, trailers, and agricultural vehicles. Characterized by diverse object categories, significant scale variations, and complex backgrounds, this dataset effectively validates our model.

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TABLE I
COMPARISON OF PERFORMANCES ON M3FD DATASET (RESULTS WITH CITATIONS ARE CITED FROM ORIGINAL PAPERS; OTHERS ARE SELF-REPRODUCED)

Model	Data	Bus	Car	Lam	Mot	Peo	Tru	mAP@0.5
YOLOv5	Thermal	78.4	88.7	70.2	63.7	80.8	65.5	74.6
YOLOv5	Visible	78.1	90.9	86.4	70.0	70.6	70.4	77.7
UMFusion[4]	RGB+T	79.5	92.0	85.6	73.1	81.8	69.5	80.3
DeFusion[44]	RGB+T	82.8	92.3	87.7	69.2	80.6	71.3	80.7
ReCoNet[45]	RGB+T	78.6	91.7	87.4	69.3	79.0	69.9	79.3
TarDAL[11]	RGB+T	81.4	91.8	87.2	69.5	81.6	68.7	80.5
LLRNet[46]	RGB+T	80.6	91.6	**88.4**	68.5	80.3	69.9	79.2
IGNet[47]	RGB+T	82.4	92.8	86.6	71.7	81.6	72.2	81.1
CDDFuse[22]	RGB+T	82.6	92.5	86.9	71.6	81.6	71.5	81.2
PSFusion[48]	RGB+T	82.0	92.3	87.1	70.9	**82.0**	73.0	81.2
Text-IF[49]	RGB+T	84.8	**94.2**	86.5	70.5	81.2	76.4	82.3
MV-MoE(ours)	RGB+T	**92.0**	91.5	75.8	**84.0**	81.9	**83.5**	**84.8**

3) M3FD [11]: This publicly available dataset consists of 4200 calibrated infrared-visible image pairs, representing four main categories and ten subscenarios, specifically designed for image fusion and object detection tasks. The images were captured on urban roads under diverse weather conditions and environmental settings. Moreover, the dataset includes manually annotated labels for 34 407 instances spanning six categories, predominantly featuring common urban road objects like pedestrians and vehicles. This dataset is therefore exceptionally well-suited for research on IR and RGB image fusion, as well as detection algorithms, in complex urban environments.

B. Settings

The models in this article were trained on NVIDIA GeForce RTX 3090 GPUs. Implemented in the PyTorch framework, training employed a stochastic gradient descent (SGD)-based optimizer with an initial learning rate of 1e-2. All models were trained for 100 epochs with a batch size of eight. The loss function applied was a multiobjective united loss, which included an object detection composite loss, expert load balancing regularization loss, and expert collaboration loss. During training, the noise perturbation module for gated paths was enabled to enhance path selection robustness. In the MoE model, a sparse expert activation strategy with $G_k = 2$ and $E_k = 2$ was employed based on preliminary experiments. Experts and groups were activated hierarchically, with independent gating weights set for both intragroup and intergroup interactions to support path-level scheduling.

C. Evaluation Metrics

To evaluate the practical performance of fused images in downstream object detection tasks, this article adopts mean average precision (mAP) as the evaluation metric. This metric comprehensively measures the model's detection performance across different categories and confidence thresholds, representing the most mainstream and widely recognized method for assessing object detection accuracy.

Specifically, for each object category, precision and recall are first calculated based on the intersection over union (IoU) between predicted and ground-truth bounding boxes. A precision–recall (PR) curve is then plotted, and the area under this curve represents the average precision (AP) for that category. The mAP is obtained by averaging the AP values across all categories

$$\mathrm{m}AP = \frac{1}{C}\sum_{c=1}^{C} \mathrm{AP}_c \tag{12}$$

$$\begin{aligned} \mathrm{AP} &= \int_0^1 P_c(r)\,\mathrm{d}r \\ &= \int_0^1 \mathrm{Precision}\;\mathrm{d}(\mathrm{Recall}) \\ &= \int_0^1 \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}\mathrm{d}\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \end{aligned} \tag{13}$$

where C denotes the number of classes in the detection task, and AP_c represents the average precision of the cth class target. For binary classification tasks (e.g., pedestrian/nonpedestrian), mAP corresponds to the average precision(AP) of the positive class. In practical experiments, we employ an IoU $\geq$ 0.5 threshold as the criterion for positive samples (mAP@0.5). Simultaneously, we utilize the more stringent mAP@0.5:0.95 to evaluate the model's comprehensive capability across varying precision requirements. By conducting object detection training and testing on fused images, mAP effectively reflects how fusion quality impacts downstream task performance, thereby providing task-level quantitative validation for the efficacy of fusion strategies.

D. Experimental Results and Analysis

Table I demonstrates the performance advantage of our MV-MoE approach on the M3FD dataset, achieving an mAP@0.5 score of 84.8% that surpasses all other methods. Compared to the best-performing single-modal method, visible, our multimodal approach achieves a 7.1% improvement in mAP@0.5, showcasing the powerful capabilities of multimodal information fusion. Furthermore, when compared to

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TABLE II
COMPARISON OF PERFORMANCES ON LLVIP DATASET (RESULTS WITH CITATIONS ARE CITED FROM ORIGINAL PAPERS; OTHERS ARE SELF-REPRODUCED)

Model	Data	mAP@0.5	mAP@0.75	mAP@0.5:0.95
Faster R-CNN[50]	RGB	91.4	48.0	49.2
Faster R-CNN	Thermal	96.1	68.5	61.1
SSD[51]	RGB	82.6	31.8	39.8
SSD	Thermal	90.2	57.9	53.5
YOLOv5	RGB	90.8	51.9	50.0
YOLOv5	Thermal	94.6	72.2	**61.9**
MV-MoE (ours)	RGB+T	**96.3**	**69.7**	61.7

TABLE III
COMPARISON OF PERFORMANCES ON VEDAI DATASET (RESULTS WITH CITATIONS ARE CITED FROM ORIGINAL PAPERS; OTHERS ARE SELF-REPRODUCED)

Model	Data	mAP@0.5	mAP@0.5:0.95
FasterR-CNN[50]	RGB	63.5	39.4
FasterR-CNN	Thermal	72.2	42.6
YOLOv5	RGB	74.3	46.2
YOLOv5	Thermal	74.0	46.1
SuperYOLO[52]	RGB+T	76.5	47.6
ICAFusion[13]	RGB+T	76.6	44.9
CMAFF[26]	RGB+T	76.9	46.6
MMYNet[53]	RGB+T	80.0	52.1
MV-MoE(ours)	RGB+T	**84.6**	**55.9**

the best-performing multimodal method, Text-IF, which excels in certain categories like Car and Lam, our method achieves higher accuracy in critical categories such as Bus and Mot, while also slightly outperforming Text-IF in overall mAP@0.5. This reveals that MV-MoE possesses distinct advantages in integrating diverse modal data, enabling stable performance across multiple key categories. However, we acknowledge that the lower performance on the "Lam" category compared to several competing methods indicates a limitation in handling certain extremely small or distant targets. A detailed investigation and discussion of this limitation are provided in Section IV-G.

Table II demonstrates that the proposed MV-MoE fusion model achieves superior comprehensive performance across three metrics on the LLVIP dataset. On mAP@0.5, MV-MoE achieved 96.3%, surpassing all comparison models, demonstrating its outstanding performance in detection recall. On the mAP@0.5:0.95 metric, MV-MoE reached 61.7%, exceeding our baseline bimodal model, YOLOv5, indicating that the fusion model exhibits greater advantages in overall balance and stability of object detection quality. Although MV-MoE scored 69.7% on the mAP@0.75 accuracy metric, it remains lower than YOLOv5-Thermal's 72.2%.

Table III demonstrates the performance comparison of our approach against other methods on the VEDAI dataset for vehicle detection. As shown in the table, the multimodal fusion network outperforms the monomodal baseline overall, and when compared with existing fusion methods, MMYNet achieved relatively good performance, with an mAP@0.5 of 80.0% and an mAP@0.5:0.95 of 52.1%. Compared to this robust model, our fusion method further improved the mAP@0.5 to 84.6% (an increase of 4.6%) and the mAP@0.5:0.95 to 55.9% (an increase of 3.8%). The experimental results demonstrate that MV-MoE's hierarchical gating and parallel expert mechanisms can effectively filter out background noise and adaptively fuse multimodal features, thereby achieving more stable and robust detection performance.

TABLE IV
ABLATION STUDIES ON THREE INFRARED-VISIBLE DATASETS

Dataset	Model	mAP@0.5	mAP@0.75	mAP@0.5:0.95
LLVIP	Baseline	95.8	71.4	62.3
LLVIP	MV-MoE	96.3 (↑0.5)	69.7 (↓1.7)	61.7 (↓0.6)
VEDAI	Baseline	79.7	47.7	46.8
VEDAI	MV-MoE	84.6 (↑4.9)	69.4 (↑21.7)	55.9 (↑9.1)
M3FD	Baseline	83.6	55.9	52.1
M3FD	MV-MoE	84.8 (↑1.2)	56.8 (↑0.9)	53.6 (↑1.5)

E. Ablation Studies

In Table IV, to validate the effectiveness of MV-MoE, we conducted ablation experiments on three infrared-visible datasets (LLVIP, VEDAI, and M3FD) by removing the MV-MoE module from the baseline model. Information exchange between the two modalities was disabled, and object detection was performed by simply summing the features from both modalities at the detector output stage. This evaluated the performance improvement of the MV-MoE model relative to the baseline model.

On the LLVIP dataset, the baseline model achieved an mAP@0.5 of 95.8%, while the MV-MoE model improved mAP@0.5%–96.3%, representing a 0.5% increase. However, mAP@0.75 and mAP@0.5:0.95 decreased by 1.7% and 0.6%, respectively. This indicates that the MV-MoE model performs better under a more lenient threshold (mAP@0.5) but shows a slight decline under stricter thresholds (mAP@0.75 and mAP@0.5:0.95).

On the VEDAI dataset, the baseline model achieved an mAP@0.5 of 79.7%, while the MV-MoE model significantly improved this metric to 84.6%, representing a 4.9% increase. Furthermore, mAP@0.75 and mAP@0.5:0.95 increased by 21.7% and 9.1%, respectively. This demonstrates a comprehensive performance improvement of the MV-MoE model on the VEDAI dataset, particularly under stricter threshold conditions.

On the M3FD dataset, the base model achieved an mAP@0.5 of 83.6%, while the MV-MoE model improved this to 84.8%, representing a 1.2% increase. Furthermore, it increased mAP@0.75 and mAP@0.5:0.95 by 0.9% and 1.5%, respectively. This demonstrates that the MV-MoE model also performs well on the M3FD dataset.

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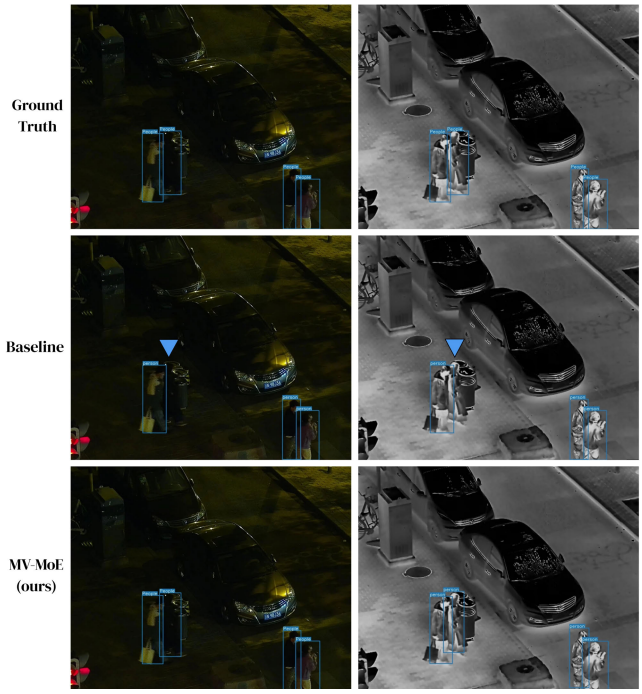


Fig. 4. Detection results comparisons on the LLVIP dataset. From top to bottom are the ground-truth annotations, the predictions of the baseline model, and the predictions of the proposed MV-MoE method. The left and right columns correspond to the RGB and IR modalities, respectively. Blue inverted triangles indicate false negatives. Compared with the baseline, MV-MoE reduces missed detections and achieves higher accuracy in multimodal detection.

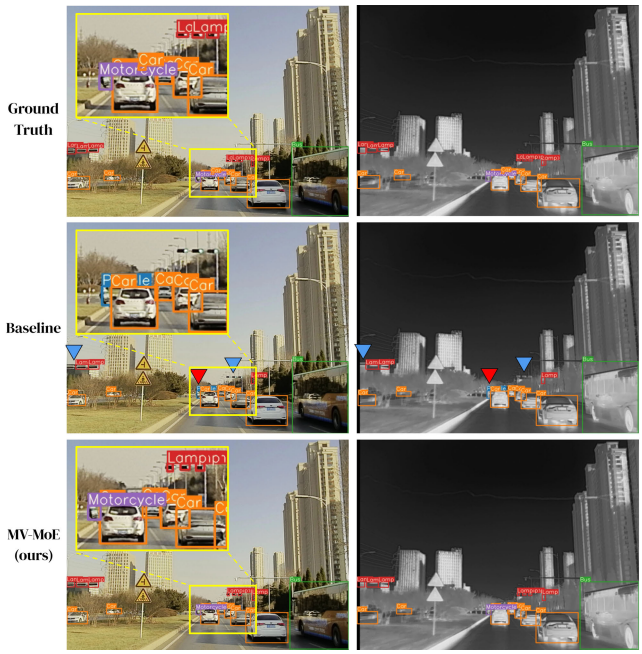


Fig. 5. Detection results comparisons on the M3FD dataset with a layout consistent with Fig. 4, respectively. Red inverted triangles indicate false positives, and blue inverted triangles indicate false negatives. Compared with the baseline, MV-MoE successfully identified missed small targets through hierarchical feature selection and a multiexpert complementary mechanism.

Additionally, Figs. 4–6 illustrate the practical detection performance of our method versus the baseline model on the LLVIP, M3FD, and VEDAI datasets. Results show that

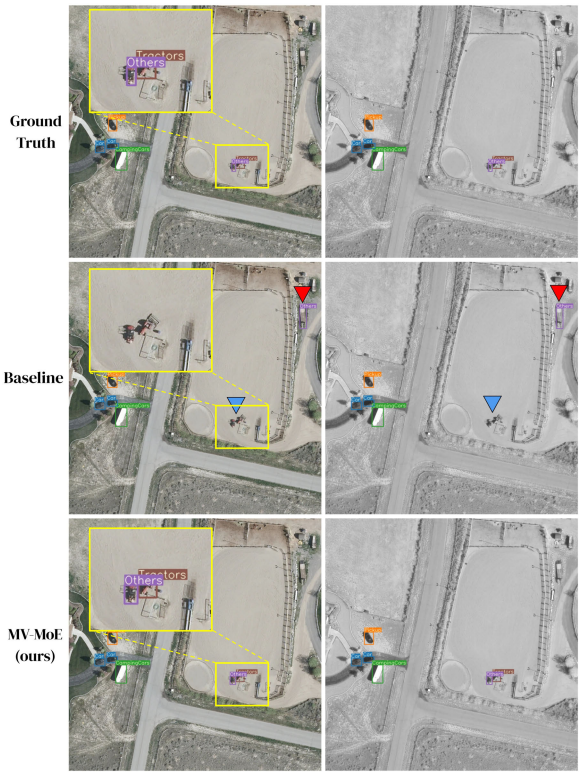


Fig. 6. Detection results comparisons on the VEDAI dataset with a layout consistent with Fig. 4, respectively. Red inverted triangles indicate false positives, and blue inverted triangles indicate false negatives. Compared with the baseline, MV-MoE reduces the false-positive rate and recovers small targets that were missed.

the baseline model still suffers from false positives and false negatives when identifying objects with colors similar to the background or with blurred features. In contrast, our method demonstrates superior performance.

Fig. 4 illustrates the detection results comparison on the LLVIP dataset. It is evident that under conditions where target contours closely match the background, and multiple individuals stand in close proximity, the RGB modality suffers from insufficient illumination and noise, while the IR modality is affected by interference from adjacent targets. The baseline model exhibits significant missed detections in the images, failing to identify human targets completely obscured in darkness and overlapping with other objects. In contrast, MV-MoE delivers more stable detection results. Despite extremely low brightness and limited visual details in the images, MV-MoE successfully detects all pedestrian targets, effectively addressing the false-negative issue observed in the baseline model.

Fig. 5 illustrates detection performance comparisons on the M3FD dataset. It is evident that the baseline model exhibits false positives and false negatives when detecting small objects in images, failing to identify distant small targets (e.g., the model fails to recognize the lamp in the distance). This phenomenon is common in cross-modal fusion tasks, particularly pronounced in scenes with strong texture interference or less prominent heat sources. In contrast, MV-MoE delivers more stable detection results. In the images, MV-MoE effectively reduces false activations in background regions,

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thereby minimizing false positives. Simultaneously, through its hierarchical feature selection and multiexpert complementary mechanism, the model concentrates responses more effectively within target regions, successfully identifying small objects missed by the baseline model.

Fig. 6 illustrates detection performance comparisons on the VEDAI dataset. This dataset features aerial remote sensing scenarios characterized by small, sparsely distributed targets against complex background textures. In IR imagery, thermal radiation from roads, bare ground, and other areas can even exceed that of target vehicles, resulting in weak contrast for vehicles in the IR modality. It can be observed that the baseline model exhibits a certain degree of detection errors. As shown in the figure, the baseline model misdetects ground structures or shadowed areas while failing to detect vehicles with weakened textures. In contrast, MV-MoE's hierarchical selection mechanism dynamically fuses expert paths suited for small and weakly contrasted targets, enhancing vehicle features. Simultaneously, shared experts provide cross-modal consistent semantic References, successfully suppressing pseudo-activation points caused by high ground temperatures, thereby reducing interference from IR features.

Our approach demonstrates stable cross-modal fusion capabilities across diverse datasets and imaging conditions. Through hierarchical selection, complementary expertise, and cross-modal consistency modeling, the module effectively reduces false positives and false negatives, achieving more reliable multimodal object detection.

In summary, the proposed MV-MoE framework exhibits robust generalization across various complex conditions. Under varying visual and shape complexities, such as extreme low-light environments and severely overlapping target shapes (e.g., the LLVIP dataset), the multiexpert complementary mechanism effectively separates target semantics from background noise. When encountering high scene complexity and thermal contrast variability—particularly in aerial scenarios (e.g., the VEDAI dataset) where intense ground thermal radiation occasionally overwhelms target signatures—the hierarchical selection successfully suppresses pseudo-activation points to maintain robust detection. Finally, regarding object size, while the dynamic fusion significantly enhances the weak spatial features of small objects in complex urban traffic (e.g., the M3FD dataset), extreme cases of distant targets coupled with single-modal signal loss still pose a challenge, which is thoroughly investigated in our subsequent failure case analysis.

F. Experiment Heatmap Analysis

Based on the heatmaps of different modality data shown in Fig. 7, we can clearly observe that the areas of interest for different modality data exhibit differences under varying lighting conditions, driven by the characteristics of each data type. For RGB images, the heatmaps reveal that the model can capture objects with higher reflectance. Under well-lit conditions, the model can identify and capture rich object features. For IR images, heat accumulation correlates with an object's thermal radiation intensity, a phenomenon unaffected by RGB conditions.

In conventional image fusion, RGB features from well-lit RGB are typically dominant, while IR features prevail in low-light conditions. However, this dominance is not fixed in complex recognition environments. As illustrated, in well-lit scenes, target feature extraction remains influenced by other highly reflective objects, causing heat distributions to nearly cover the entire image. In some cases, heat concentration in nontarget areas (e.g., ground and vehicles) within single-modal data even exceeds the total heat concentration of the target region in multimodal data.

For such scenarios, we should suppress the dominance of these features and instead prioritize IR image features with superior aggregation effects in fusion. The dynamic fusion method proposed in this article serves as an effective solution. This approach suppresses nontarget features while enhancing beneficial ones, thereby improving the quality of the fused image.

To further investigate the effectiveness of the dynamic fusion strategy, Fig. 7 visualizes the feature maps augmented with a continuous color scale for response intensity (ranging from low/blue to high/red) and numbered bounding boxes to track specific target regions.

In the daytime scenarios, while the RGB modality captures rich visual details, its prefusion feature maps often exhibit scattered high responses caused by background textures and reflections (e.g., the scattered activations around Box 1 and Box 3). Meanwhile, the IR modality focuses on thermal signatures (Box 2 and Box 4). After applying MV-MoE, the background interference in the RGB branch is noticeably mitigated, and the high-response regions in both modalities become more concentrated on the valid targets (e.g., pedestrians and vehicles within the red dashed boxes).

A similar trend is observable in the nighttime scenes. The widespread background noise caused by ambient light in the prefusion RGB maps (e.g., Box 5 and Box 7) is visibly reduced after fusion. More importantly, the visualizations suggest an improved cross-modal feature alignment. Prior to fusion, the IR and RGB feature maps exhibit spatial discrepancies in their attention regions. Following the hierarchical routing of MV-MoE, the high-response areas of both modalities tend to converge more consistently within the corresponding target bounding boxes (e.g., comparing the aligned responses in postfusion Box 5 and Box 6, or Box 8 and Box 9). This convergence indicates that the module helps align cross-modal semantics and adaptively adjusts modal reliance based on regional reliability. This visually observed enhancement in semantic consistency provides a reasonable visual explanation for the framework's overall effectiveness.

G. Failure Case Analysis and Discussion

To provide a comprehensive and objective evaluation, we conducted an in-depth analysis of categories that exhibited significant performance degradation in the quantitative experiments on the M3FD dataset (such as the "Lam" category). As shown in the actual detection visualizations in Fig. 8, although the model was able to accurately detect most subject targets, it exhibited significant missed detections for certain specific targets.

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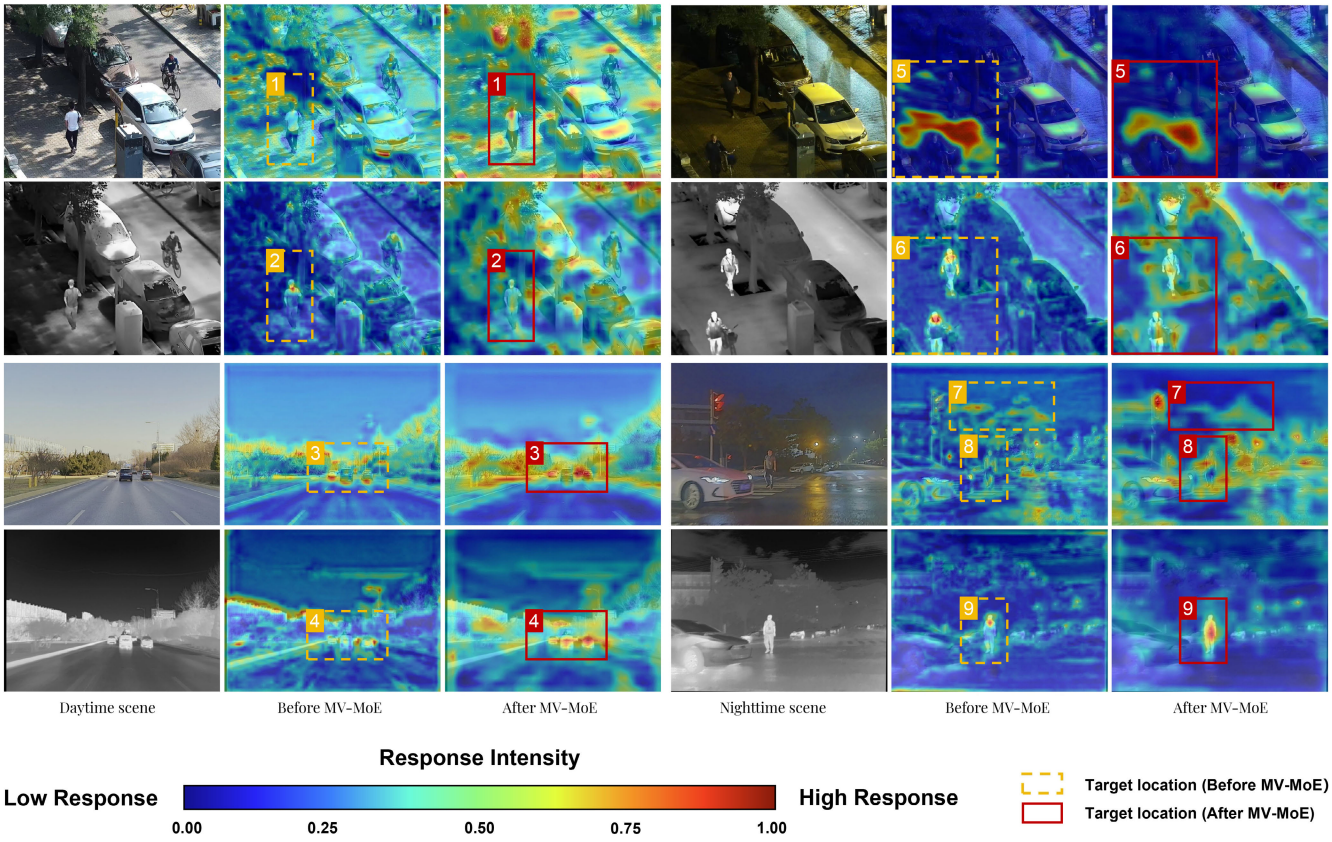


Fig. 7. Visualization of feature prediction process. The figure displays heatmap distributions of RGB and IR images before and after MV-MoE processing in daytime and nighttime scenarios. The left and right sides of the figure correspond to daytime and nighttime scenes, respectively. Rows 1 and 3 display RGB images and their feature maps, while rows 2 and 4 show IR images and their feature maps. Columns 2 and 4 present input image features before module processing, and columns 3 and 6 present input image features after MV-MoE processing.

By carefully comparing the ground truth labels and analyzing these missed targets, we found that the core cause of the performance decline was not the category itself, but rather that these missed targets exhibited highly consistent characteristics, such as being located at the edges of the image, situated at the farthest reaches of the field of view, or having extremely weak feature information in a single modality. This phenomenon indicates that the model still faces certain general limitations in its detection and perception capabilities for extremely small targets with such characteristics. Based on the common characteristics of the aforementioned missed targets, we combined the model's feature extraction and adaptive data fusion mechanisms to conduct an in-depth analysis of the underlying causes.

In terms of spatial feature representation, very small objects—such as edges or distant targets—occupy very few spatial pixels. During the downsampling process in the deep layers of the baseline network, the weak and fine-grained spatial features of these objects are easily diluted to the point of being overwhelmed by the intense background noise in the surrounding area.

In addition to the inherent fragility of monomodal features, such minute targets also face severe information asymmetry during the multimodal fusion stage. Take the distant streetlight in Fig. 8 as an example: although the RGB image retains its faint but discernible outline, due to the physical characteristics of IR data acquisition, these targets lack significant thermal radiation features, resulting in the near-complete loss of their raw signals in the IR modality. When the IR branch passes these highly degraded semantic features—or pure background noise—to the MV-MoE module, this extreme information asymmetry disrupts the balance of the dynamic routing mechanism. At this point, severe "negative transfer" occurs, ultimately leading to spatial shifts in the predicted bounding boxes or complete missed detections.

The performance degradation in the "Lam" category objectively reflects the common challenges faced by multimodal dynamic fusion networks when processing small targets at the edges, in the distance, or where modal information is severely imbalanced. This limitation highlights the difficulty of preserving weak cross-modal features under strong background noise interference and points the way toward exploring more robust fusion strategies for extremely small targets in the future.

H. Computational Complexity Analysis

To evaluate deployment efficiency, we benchmarked representative models on a single RTX 3090 GPU. As shown in Table V, the monomodal YOLOv5l achieved the highest speed (78.52 FPS). Lightweight multimodal networks such as CMAFF and MMYNet, which have fewer parameters, maintained inference speeds of around 46 to 51 FPS. In contrast, to

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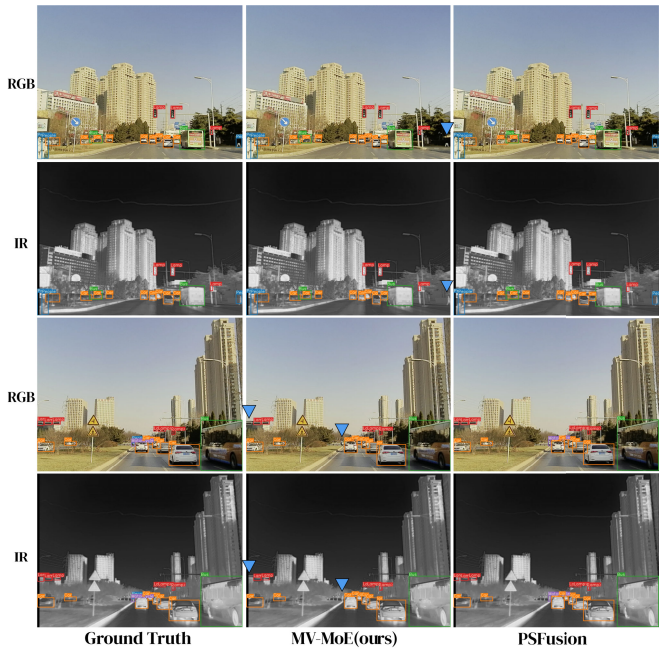


Fig. 8. Visual comparison of typical false negatives in the M3FD dataset. This figure illustrates the actual detection performance of our model compared to other models in an urban setting. The first column shows the ground truth, the second column shows our method, and the third column shows the comparison method, PSFusion. The blue inverted triangles in the figure indicate instances of false negatives detected.

TABLE V
COMPARISON OF PERFORMANCE AND COMPUTATIONAL COMPLEXITY AMONG REPRESENTATIVE MODELS ON THE VEDAI DATASET

Model	mAP@0.5	GFLOPs	Params(M)	FPS
YOLOv5l	74.3	36.7	46.6	78.52
CMAFF[7]	76.9	42.2	16.48	46.38
MMYNet[53]	80.0	43.0	17.16	51.27
MV-MoE(ours)	84.6	133.1	94.38	31.46

handle complex multiview features, our MV-MoE introduces a heavier dual-stream backbone and multiple expert modules, resulting in relatively high parameter counts (94.38 M) and computational overhead (133.1 GFLOPs). However, thanks to MoE's sparse gating mechanism, the network activates only a subset of experts during inference. Consequently, MV-MoE still maintains a processing speed of 31.46 FPS. While MV-MoE does not excel in terms of extreme lightweight design and inference speed, it still meets basic real-time requirements (>30 FPS) while achieving a detection accuracy 4.6% higher than MMYNet (mAP@0.5). This indicates that MV-MoE offers a tradeoff between detection accuracy and computational cost.

V. CONCLUSION

This article proposes a multimodal image fusion framework named MV-MoE, based on a parallel expert group and hierarchical gating mechanism. This framework addresses limitations in existing infrared-visible fusion methods, such as structural monotony, shallow fusion paths, and uneven expert scheduling. Experimental results demonstrate that MV-MoE outperforms mainstream monomodal and multimodal detection methods across multiple typical fusion datasets. While ensuring prominent representation of IR targets, the model demonstrates robust fusion capabilities and perceptual acuity in both high IoU precision matching and overall detection performance. Further analysis reveals that the designed framework effectively mitigates multimodal redundancy coupling and information conflicts, enhances the discriminative power of fused features, and improves adaptability to downstream tasks. Future work will explore extending this framework to other multimodal combinations (e.g., RGB-Depth and RGB-LiDAR) to evaluate its generalization across diverse object categories and domains. Concurrently, we will further optimize the information flow mechanism and fusion granularity among experts to achieve a more efficient and generalizable intelligent perception system.

It is worth emphasizing that while MV-MoE takes fused RGB and IR images as input, it does not merely stack the two modalities. Instead, it constructs multiview semantic paths through parallel expert modules and dynamically schedules fusion strategies under a hierarchical gating mechanism, thereby effectively capturing complementary information between the two modalities. Compared to conventional early fusion methods, MV-MoE mitigates modal interference and redundancy conflicts, enhancing the accuracy and robustness of downstream object detection.

However, although the overall performance has been effectively validated, the current model's ability to detect small objects in complex backgrounds remains limited. The current limitations stem from the fact that fine-scale features are easily lost during downsampling in the deep layers of the network and are highly susceptible to background noise. To address this, we will pursue improvements in two directions. First, we plan to upgrade the underlying feature extraction architecture by adopting structurally optimized modules. Combining these with high-resolution shallow features will enhance the model's ability to capture and retain extremely small targets. Second, we aim to introduce a heterogeneous expert collaboration mechanism. By designing expert modules with clearly defined roles—such as employing frequency-domain experts for high-frequency detail extraction and large-kernel convolutional experts for global spatial modeling—we can independently handle cross-modal information differences. This approach enables a more interpretable and reliable multimodal perception framework.

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